WRIGHT STATE UNIVERSITY

COLLEGE OF ENGINEERING AND COMPUTER SCIENCE

Ethics of Data Science and AI

Week 11 Thu: bias, privacy law, AI



Ethics: telling right from wrong with data

Business	Who gets a loan, a job interview, a discount
Health	Who gets flagged for extra care
Transport	Routes, safety systems, self-driving
Civics	Policing, benefits, school admissions

Data decisions change real lives. Ethics asks: is this decision fair, lawful and honest?

Where bias comes from

Source	What goes wrong	Example
Training data	Past unfairness is learned	Mostly male CEOs in the data: the model repeats it
Who is collected	Some groups are missing	A survey that only some people answer
Prompt wording	Assumptions steer the answer	Adding a demographic cue changes the output
Sycophancy	The tool agrees with you	A chatbot confirms your biased idea

A model is not neutral. It reflects the data it was given.

Example: what image generators show us

Who appears

Men about 3 times as often as women.

Which jobs

High-paying jobs: mostly lighter skin. Low-paying jobs: mostly darker skin.

"Politician", "lawyer", "judge", "CEO" produced almost entirely light-skinned men.

Bloomberg test of Stable Diffusion, 2023, reported by the University of Michigan.

Bias and fairness are not the same word

Bias

An unfair tilt in the data, the model, or the result.

Fairness

The goal we check against. Say which meaning you use.

Check	Question
Presence	Is every group in the data?
Errors	Does the model make the same mistakes for every group?

Bias in lending, medicine and policing

Field	What happened	Lesson
Lending, insurance	Models learn from past approvals	Past unfairness gets copied
Medicine	A hospital algorithm used past spending as a stand-in for need (2019 study)	The wrong target column hides bias
Policing	COMPAS risk score: more false "high risk" labels for Black defendants (ProPublica, 2016)	Error rates differ by group
Hiring	An experimental recruiting tool downgraded resumes mentioning women (Reuters, 2018)	Learning from history repeats history

Removing a column does not remove bias

1 **Sensitive column: Gender, Race, Age**

2 **Delete it from the model**

3 **Proxy column: zip code, name, school**

4 **Bias can still come back through the proxy**

What to do

Keep the sensitive column to check results by group.

Ask of every feature: could this stand in for it?

Spot the bias: three cases

Case	Where is the bias?
A loan model trained on past approvals uses zip code.	Unfair history in the data, plus a proxy column
An app rates its own fairness from voluntary reviews.	Voluntary response: only some people are in the data
A face tool tested mostly on one group fails on others.	A group is missing from the data

For each case, name where the bias entered and one fix.

How to reduce bias

1 Look	Who is in the data? Who is missing?
2 Split	Compare results for each group
3 Document	Write down limits and choices
4 Review	A person checks decisions that matter

If it cannot be made fair, the ethical choice may be not to use it.

Personal data: what the law protects

Direct

Name, email, ID numbers.

Indirect

Location, device, face, voice, birth date plus zip code.

Sensitive

Health, money, children, biometrics.

Combining harmless columns can identify a person.

Laws that protect personal data

Law	Where	What it covers
GDPR	European Union	Data about people in the EU; consent, access, deletion; fines up to 4% of turnover
CCPA / CPRA	California	Right to know, delete, and opt out of sale
HIPAA	United States	Health information held by providers and insurers
FERPA	United States	Student education records
COPPA	United States	Data from children under 13
Ohio	Ohio	Breach notice; 2018 safe harbor; no full privacy law as of May 2026

Four ideas that privacy laws share

Consent	Say what you collect and why
Minimum	Collect only what you need
Rights	See, correct, delete my data
Security	Protect it. Tell people if it leaks

If a column is not needed, drop it. It also lowers the privacy risk.

Generative AI and your data

Risk	What to do
Prompts may be stored or used for training	Read the privacy policy and account settings
48% of employees surveyed entered non-public company data (Cisco, 2024)	Never paste personal or confidential data
Vendors are vague about training data	Research the tool and company first

Assume the provider can see what you type.

Wrong answers and fake media

Hallucination

Confident, fluent, and wrong. Sources can be invented or misdescribed.

Verify every claim.

Deepfakes

Fake media of a real person, made without consent. Used to harass and mislead.

Disclose AI use.

Check a claim against a real source before you share it.

Copyright and your own work

Issue	What to do
Who owns AI output? - ***Unsettled***	Do not assume it is free to publish
Uploading copyrighted material	Get permission first
Submitting AI work as yours	That is misconduct. Disclose what you used
Course rule	AI may help with code. The analysis must be yours

Hidden costs: energy, labor and access

Issue	What the guides say
Environment	Large models use electricity and cooling water; the full impact is hard to measure
Human labor	Labeling and moderation work is often low-paid and outsourced; workers may see disturbing content
Equity and access	Less common languages are underrepresented; cost and internet access limit use
Benefit to weigh	Accessibility help: simpler text, alt text, step-by-step plans

Researchers mapped 19 areas of concern

Area group	Includes
Fairness and safety	Fairness and bias, safety, alignment, evaluation and auditing
Content and trust	Harmful content, hallucinations, transparency, interaction risks
Data and law	Privacy, security, copyright, governance and regulation
Society	Labor and economy, education, art, writing and research, sustainability

378 issues, 19 areas (Hagendorff, 2024). We covered most of them today.

Five questions before you use a tool

1. Is the benefit worth the energy it uses?
2. Does uploading this content raise copyright or privacy issues?
3. What data does the tool keep about me?
4. How will I check that the answer is true?
5. How will I check the output for bias?

Discussion: which law, which bias?

Case	Ask
A clinic uses an AI tool to flag patients for follow-up.	Which law? Whose data is missing?
A kids' game collects the birthdays of players aged 10.	Which law applies?
A US company trains a model on customers in Germany.	Which law applies?

Name the law, name the bias, name one fix.

Sources for this lecture

Source	Used for
University of Michigan, Ethical Considerations of Using GenAI Tools (2025)	Bias, data, copyright, environment, equity, tips
Hagendorff (2024), Mapping the Ethics of Generative AI, MIT AI Risk Repository	19 areas and 378 issues
University of Alberta Library, GenAI Ethics guide	Labor, accessibility, integrity, questions to ask
Privacy laws in Ohio overview (privacyon.com, May 2026)	Ohio privacy status